CODING & SOLUTION

Project Title

Rising Waters

Project Subtitle

AI-Powered Flood Prediction System using Machine Learning

Introduction

Rising Waters is an AI-powered Flood Prediction System developed using Python and Machine Learning. The application predicts flood risk by analyzing rainfall and environmental parameters. The project provides a simple, accurate, and user-friendly web application for disaster management.

Programming Language

The application is developed using Python because it provides powerful libraries for data analysis, machine learning, and web application development.

Technologies Used

- Python
- Streamlit
- Pandas
- NumPy
- Scikit-learn
- XGBoost
- Joblib
- GitHub

Machine Learning Model

The project uses the XGBoost algorithm for flood prediction. The model is trained using historical rainfall and weather data and provides high prediction accuracy.

Solution Workflow

Step 1

The user opens the Rising Waters application.

Step 2

The user enters rainfall and environmental parameters.

Step 3

The application validates the input data.

Step 4

The trained XGBoost model processes the data.

Step 5

The model predicts the flood risk.

Step 6

The prediction result is displayed on the Streamlit dashboard.

Coding Standards Followed

• Modular programming
• Proper indentation
• Meaningful variable names
• Code comments
• Error handling
• Reusable functions

Advantages

• Accurate prediction
• Fast processing
• Easy maintenance
• User-friendly interface
• Scalable application
• Reliable results

Expected Outcome

The Rising Waters application provides quick and reliable flood predictions, helping authorities issue early warnings and improve disaster preparedness.

Conclusion

The coding implementation follows software engineering best practices and Machine Learning techniques to deliver an efficient and reliable flood prediction system. The application is easy to maintain, scalable, and suitable for real-world deployment.